

Topic 3: Healthcare Data Standards — FHIR R4, LOINC, & RxNorm

TOPIC OBJECTIVE

Healthcare data is historically fragmented across incompatible vendor formats. HL7 FHIR R4 solves this by defining standardized JSON resources. AegisClinical consumes FHIR bundles and decodes universal medical ontologies: LOINC for laboratory observations, RxNorm for active medications, and ICD-10 for chronic conditions, transforming raw records into verifiable clinical vectors.

1. The Interoperability Crisis: Why FHIR R4 Matters

For decades, hospital Electronic Health Records (EHRs) were walled gardens. Epic used Chronicles, Cerner used Millennium, and each hospital used idiosyncratic database schemas. A blood test for potassium might be stored as 'POTASS' in one clinic, 'K+' in another, and 'Serum Potassium' in a third.

Fast Healthcare Interoperability Resources (HL7 FHIR, pronounced 'fire') solved this problem by introducing modern, REST-friendly, resource-oriented JSON architectures. Release 4 (FHIR R4) is now the legally mandated interoperability baseline for healthcare software in the United States and Europe.

2. The Anatomy of a FHIR R4 Bundle

In FHIR, all clinical data is modeled as **Resources**. A patient's complete chart is packaged inside a Bundle containing a list of discrete entry objects:

FHIR Resource	Clinical Meaning	Key Fields Parsed by AegisClinical
Patient	Demographics and patient identity	id, name.given, name.family, gender, birthDate
Observation	Laboratory tests, vital signs, physical exam	code.coding[].code (LOINC), valueQuantity.value, unit, effectiveDateTime
MedicationRequest	Prescription order / active medication	medicationCodeableConcept.coding[].code (RxNorm), text, authoredOn
Condition	Diagnosed disease / active problem list	code.coding[].code (ICD-10/SNOMED), clinicalStatus, text

3. The Three Universal Ontologies Used in AegisClinical

To prevent ambiguity, AegisClinical does not rely on free-text pattern matching. It decodes standard healthcare ontologies:

A. LOINC (Logical Observation Identifiers Names and Codes)

LOINC is the global standard for identifying health measurements, observations, and laboratory tests. AegisClinical explicitly monitors these LOINC identifiers:

LOINC Code	Analyte	Unit	Normal Range
2160-0	Serum Creatinine	mg/dL	0.7 – 1.3 mg/dL
2823-3	Serum Potassium	mEq/L	3.5 – 5.0 mEq/L
718-7	Hemoglobin	g/dL	12.0 – 17.5 g/dL
777-3	Platelets	x10^3/uL	150 – 450 x10^3/uL
2345-7	Blood Glucose	mg/dL	70 – 140 mg/dL

B. RxNorm (Normalized Drug Nomenclature)

Maintained by the US National Library of Medicine, RxNorm provides normalized names and codes for clinical drugs. A physician might prescribe 'Prinivil', 'Zestril', or 'Lisinopril 20 MG'. RxNorm links all of these to a single concept (RxNorm: 314076), allowing AegisClinical's Pharmacology Agent to reliably map them to the **ACE-Inhibitor** class.

C. ICD-10-CM (International Classification of Diseases)

Used for medical diagnostic coding. AegisClinical inspects ICD-10 codes such as I10 (Essential Hypertension), E11.9 (Type 2 Diabetes Mellitus), and N18.3 (Chronic Kidney Disease Stage 3) to evaluate drug-disease contraindications.

4. Raw Synthea JSON Example: Serum Creatinine

```
{ "resourceType": "Observation", "status": "final", "code": { "coding": [{ "system": "http://loinc.org", "code": "2160-0", "display": "Creatinine [Mass/volume] in Serum or Plasma" }], "text": "Serum Creatinine" }, "effectiveDateTime": "2026-09-12T14:30:00Z", "valueQuantity": { "value": 2.4, "unit": "mg/dL" }, "referenceRange": [{ "low": { "value": 0.7 }, "high": { "value": 1.3 } } ] }
```

KNOWLEDGE CHECK & NEXT STEP

- 1. Why is LOINC 2160-0 used instead of searching for the word 'Creatinine'?**
Because different hospitals label creatinine differently ('Serum Cr', 'CREAT', 'Creat'). LOINC guarantees semantic precision.
- 2. What is the role of RxNorm in catching the Triple Whammy?**
It allows the system to recognize that brand names like 'Lasix' and generic 'Furosemide' belong to the same loop diuretic class.
- 3. Next Step in Topic 4: The Ingestion Parser —** How src/parser.py chronologically orders labs and builds patient vectors with zero dependencies.